

Note that I will only focus on one way of solving the problems in this document. You are welcome and encouraged to explore other methods.

Hints for Solving the Problem 1

1. **Assume the form of the polynomial**

Let $R(x) = ax^2 + bx + c$, where a, b, c are constants.

2. **Use the fact that the polynomial has three distinct zeros**

Since $R(x)$ has three distinct zeros r_1, r_2, r_3 , write the system:

$$R(r_1) = 0, \quad R(r_2) = 0, \quad R(r_3) = 0$$

3. **Form a system of equations**

Set up equations of the form $ar_i^2 + br_i + c = 0$ for each of the zeros r_1, r_2, r_3 .

4. **Express the system as a matrix equation**

Write the system as $A \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$, where A is the matrix of coefficients.

5. **Analyze the matrix for invertibility**

Determine if the matrix A is invertible by calculating its determinant.

6. **Conclude based on the invertibility of the matrix**

If the matrix is invertible, what does this imply about the solution to the system?

7. **Final conclusion**

What must the coefficients a, b, c be for the system to have only the trivial solution?

Hints for Solving the Problem 2

1. A set of functions forms a basis for P_n (the space of polynomials of degree at most n) if they are:

- **Linearly independent (LI):** No function in the set can be written as a linear combination of the others.
- **Spanning:** Any polynomial in P_n can be written as a linear combination of the functions in the set.

Why is it enough to linear independence to prove that Lagrange basis functions form a basis for P_n ?

2. To prove that $\ell_0(x), \ell_1(x), \dots, \ell_n(x)$ are linearly independent, suppose there exists a linear combination:

$$c_0\ell_0(x) + c_1\ell_1(x) + \dots + c_n\ell_n(x) = 0.$$

Use a property of Lagrange basis functions to get:

$$c_i = 0 \quad \text{for each } i = 0, 1, \dots, n.$$

What does this prove?

Hints for Solving the Problem 3

This is a computational problem.

Hints for Solving the Problem 4

The intention is for you to construct a fifth-order polynomial by hand using Lagrange interpolation. The goal is to give you a sense of how interpolation can be applied to real-world data, such as predicting future census values.

That said, as you'll quickly see, manually computing higher-order interpolations is extremely tedious and prone to numerical instability. I wanted you all to get the experience of doing it by hand first so that it will help you appreciate both the power of these methods and the necessity of coding in applied mathematics.